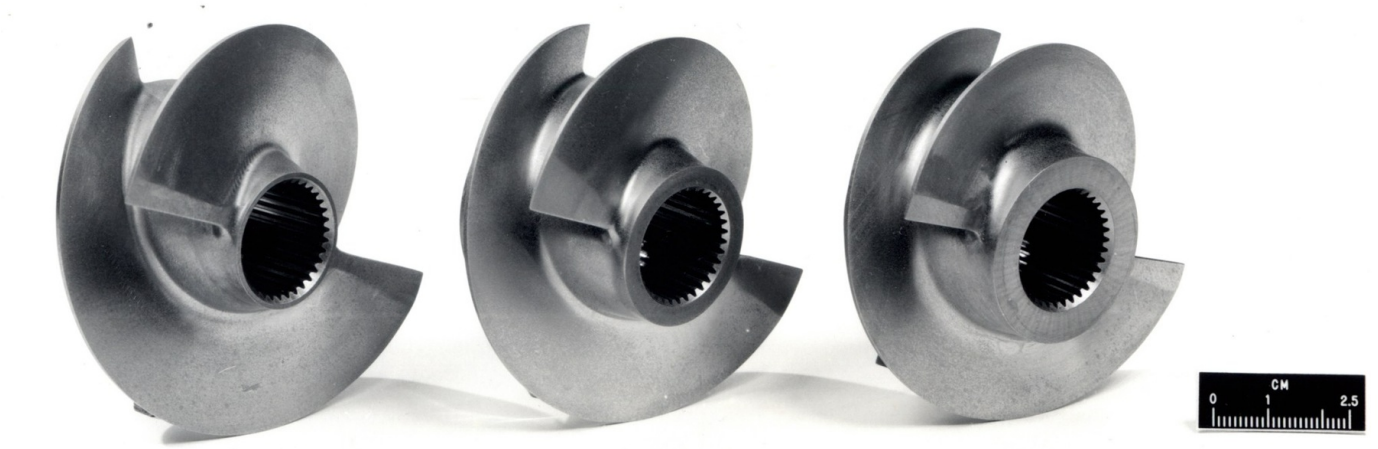
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Cavitation changes the circuit



Three liquid-rocket turbopump inducers with different blade angles, photographed at NASA Lewis Research Center in 1967. The hardware was part of a broader program studying pumps in boiling hydrogen. Photo: NASA, GRC-1967-C-03127. Source and context [1].

The pressure source that changes when you connect it

Imagine a clean pressure tone placed at a pump inlet. You know its amplitude and frequency, so the rest feels like a transmission problem: march the tone through the feed line, account for losses and reflections, and predict what reaches the tank. That is a useful first screen. It can also hide the most interesting physics.

When vapor cavities grow and collapse around an inducer, the pump inlet does not behave like a fixed source bolted to an indifferent pipe. The cavity volume responds to inlet pressure and flow. In the language of system dynamics, the inducer acquires compliance and a flow-dependent feedback term. Change the inlet-line impedance or operating point, and you may change not only what the source transmits, but how the source itself behaves.

Today’s objective is to recognize that boundary: when a one-way source model is adequate, when cavitation must enter the dynamic circuit, and what a simple lumped calculation can—and cannot—tell us about the observed frequency range.

The source is attached to a spring you did not draw

NASA's early liquid-hydrogen pump work varied inducer geometry, inlet pressure and supply-line length because the pump and feed system could not be understood independently [1,2]. Modern pump-dynamics language captures one part of that interaction with cavitation compliance. Using the sign convention in this lesson,

$$C_c = -\frac{\partial V_c}{\partial p_i}\bigg|_{\dot{m}} > 0$$

Here V_c is cavity volume and p_i is inducer-inlet pressure. Raise the pressure and the vapor volume normally shrinks, so the minus sign makes compliance positive. Its units are m³/Pa. This is an incremental, operating-point quantity—not a universal material property.

A second parameter, commonly called the mass-flow gain factor, describes how cavity volume changes with inlet flow. Together, compliance and mass-flow gain appear in small-signal pump transfer matrices and can influence stability. Brennen's review emphasizes that both are critical to pump/system dynamics, while also noting how sparse the relevant transfer-function data remain [3].

For a first physical picture, replace the upstream line by a fluid inertance. For a uniform line of length l and area A,

$$L_f = \frac{\rho \ell}{A}$$

The line's moving liquid supplies the inertia; the cavity-volume response supplies the spring. Ignore damping, mean-flow effects, downstream coupling and wave propagation for the moment. Their lumped natural frequency is then

$$f_n = \frac{1}{2\pi\sqrt{L_f C_c}}$$

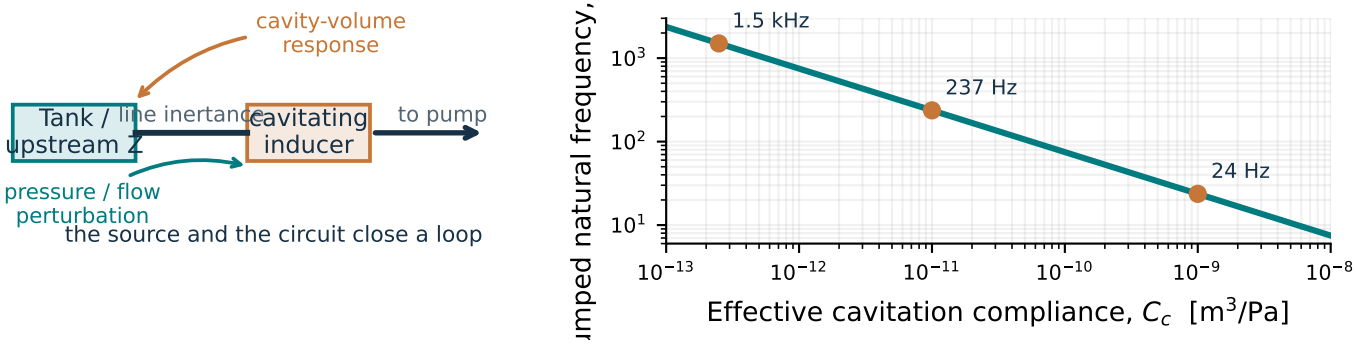


Figure 1. Original conceptual illustration and illustrative calculation. Left: a cavitating inducer closes a feedback loop with the upstream hydraulic impedance. Right: sensitivity of the ideal lumped resonance to assumed compliance for the example developed on page 3. This is not measured hardware data.

A frequency can rule out the simplest story

Take a deliberately spare cryogenic example: liquid density 71 kg/m³, a 5 m straight line and a 0.10 m inside diameter. These values are **illustrative only**; they are not telemetry, qualification limits or a reconstruction of any flight system. The flow area and line inertance are

$$A = \frac{\pi D^2}{4} = 7.85 \times 10^{-3}\ \mathrm{m}^2$$

$$L_f = \frac{(71)(5)}{7.85 \times 10^{-3}} = 4.52 \times 10^{4}\ \mathrm{Pa\,s^2/m^3}$$

Now suppose the effective cavitation compliance is 10^{-9} m³/Pa. The lumped frequency is about 23.7 Hz. Reduce that compliance by a factor of 100 and the frequency rises by only a factor of ten, to about 237 Hz. The square-root dependence is the point: large uncertainty in compliance maps into substantial—but slower—frequency movement.

$$C_c = 10^{-9}\ \mathrm{m^3/Pa} \quad \Longrightarrow \quad f_n \approx 23.7\ \mathrm{Hz}$$

$$C_c = 10^{-11}\ \mathrm{m^3/Pa} \quad \Longrightarrow \quad f_n \approx 237\ \mathrm{Hz}$$

Turn the question around. What compliance would this same lumped line need to resonate at 1.5 kHz?

$$C_c = \frac{1}{L_f (2\pi f_n)^2} \approx 2.49 \times 10^{-13}\ \mathrm{m^3/Pa}$$

That number is not proof that a 1.5 kHz observation cannot involve cavitation. It is a useful filter on the hypothesis. A high-frequency tone may instead be tied to blade passing, a higher-order cavitation pattern, a local acoustic mode, a screen or valve impedance, or structural response. NASA high-speed-video work has documented several distinct inducer cavitation forms—rotating, alternating-blade, asymmetric and higher-order—and showed that their frequencies change with inlet-pressure boundary conditions [4]. One label, “cavitation,” does not imply one oscillator.

The calculation also warns against stretching a lumped model past its wavelength limit. Once acoustic travel time along the line matters, pressure and flow are distributed states. Reflections, junctions, bends, screens, valves and two-phase regions create frequency-dependent impedance. At that point a 1-D transfer-matrix or distributed acoustic model is the natural next rung, and the pump boundary should remain coupled if source–load interaction is plausible.

A hierarchy, not a single heroic model

The practical move is not to jump immediately from a source spectrum to multiphase CFD. It is to ask what decision each model can support, then climb only when the simpler model cannot answer it.

Model rung	What it represents	Good for	It cannot establish
1. One-way source	Prescribed pressure or flow at the inlet; passive line/filter transfer	First screen of attenuation, reflection and structural forcing	Source shift, low-frequency stability or pump/load feedback
2. Coupled 1-D network	Distributed lines plus pump transfer matrix, tank and component impedances	Resonances, sensitivity, source–load interaction and stability margins	Detailed cavity topology or local three-dimensional forcing
3. Test-informed multiphysics	Unsteady cavitation, rotor geometry and structural/acoustic response	Mechanism studies and difficult local environments	Certainty without validated boundary conditions and dynamic data

Table 1. A decision-oriented modeling hierarchy. The governing program requirements and evidence needs determine the appropriate fidelity; no rung is universally mandatory.

For a one-way tone study, write the assumption plainly: the imposed source is independent of upstream impedance and operating-point perturbations over the frequency range of interest. Then test that assumption. Sweep plausible tank and line impedances. Compare the predicted inlet pressure and flow perturbations with the regime used to characterize the source. If a small load change produces a large boundary response, the source-only abstraction is not robust.

For the coupled rung, the scarce quantities are often not pipe dimensions but pump dynamics. Measure or bound compliance and mass-flow gain versus cavitation number, flow coefficient and frequency; retain phase, not only magnitude. Use pressure sensors on both sides of the inducer and enough line instrumentation to separate traveling and reflected content. If possible, vary upstream impedance deliberately. A response that moves with line length or boundary condition is powerful evidence of a coupled system mode.

NASA’s open-inducer research used an unrestricted geometry and water testing to study paths into rotating cavitation instability [5]. Earlier Fastrac work likewise connected localized rotating-cavitation behavior with system surge components and compared water-flow and liquid-oxygen observations [6]. Those programs are reminders that mechanism identification benefits from changing the circuit—not merely recording one operating point.

Finally, carry uncertainty honestly. Cavitation compliance is nonlinear and operating-point dependent; fluid properties, dissolved gas, thermal effects and scaling from water to cryogens can all matter. A tidy frequency match from one assumed compliance is a hypothesis, not validation.

What to ask when the spectrum arrives

Begin with the frequency fingerprint. Is the feature synchronous with shaft or blade order, sub-synchronous, broadband, or tied to an acoustic travel-time family? Does it move with inlet pressure, flow coefficient or tank level? Does its phase relation across sensors resemble a standing wave, a traveling wave, a structural mode or a rotating disturbance? These questions sort mechanisms before model detail obscures them.

Then separate the two safety questions. **Transmission:** can the pressure disturbance reach a sensitive tank, valve or structure with damaging amplitude? **Stability:** can the coupled pump–feed–structure loop generate or amplify that disturbance? The first can often start with measured forcing and passive transfer. The second cannot be answered without the source dynamics and feedback path.

NASA SP-8052 remains valuable public design guidance on turbopump inducers [2], but it is a 1971 design-criteria monograph, not a universal program requirement. Today’s governing contract, approved analysis plan and test requirements control. Its enduring lesson is methodological: suction performance, cavitation behavior, mechanical integrity and system dynamics have to be considered together.

Three takeaways

1. A cavitating inducer can be both a pressure source and an operating-point-dependent dynamic element.
2. The simple relation $f_n \propto (L_f C_c)^{-1/2}$ is a powerful order-of-magnitude test, not a substitute for distributed modeling or measured transfer data.
3. A source-only model is defensible when declared and sensitivity-tested; it is not a stability model.

Reflection: A narrow pressure peak moves downward when the upstream line is lengthened, while shaft speed is unchanged. What should rise on the hypothesis list?
Answer: A coupled hydraulic or acoustic mode. The line change altered inertance or travel time; a purely prescribed shaft-order source would not be expected to shift frequency with that boundary change.

Sources, scope and figure credits

[1] NASA Glenn Research Center, “Rocket Systems Area — Pumps and Tanks,” historical overview and image gallery, accessed October 10, 2026. Opening photo: GRC-1967-C-03127 (1967), NASA.

[2] J. K. Jakobsen and R. B. Keller Jr., “Liquid Rocket Engine Turbopump Inducers,” NASA SP-8052, May 1, 1971. U.S. Government work; public use permitted.

[3] C. E. Brennen, “On the Dynamics of a Cavitating Pump,” IOP Conference Series: Earth and Environmental Science 49, 052018, December 13, 2016. CC BY 3.0. DOI 10.1088/1755-1315/49/5/052018.

[4] P. Arellano et al., “Characterization of Axial Inducer Cavitation Instabilities via High Speed Video Recordings,” NASA NTRS 20120002837, presented December 5–9, 2011.

[5] C. E. Brennen et al., “A New Modeling Approach for Rotating Cavitation Instabilities in Rocket Engine Turbopumps,” NASA NTRS 20160006403, 2016.

[6] T. Zoladz, “Observations on Rotating Cavitation and Cavitation Surge from the Development of the Fastrac Engine Turbopump,” NASA NTRS 20000067660, 2000.